

Polyolefin nanocomposite fibres and films

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11.1 Introduction

Nanotechnology is concerned with the creation, application, and manipulation of materials or structures on a 1–100 nm. In the polyolefin area, the most important nanotechnology is the inclusion of nanosized materials into polyolefin matrices. Due to their extremely small dimensions, nanomaterials, also known as nanoparticles, have some extraordinary properties compared to their bulkier counterparts. The high specific surface area is one of those properties very useful in material engineering. Upon the addition of nanoparticles in a polymeric material, the resultant material, also called nanocomposite, can have some performance improvements which are not achievable by using the polymer alone. For example, adding nanoclay to polypropylene (PP) can make the modified propylene dyeable with acid and disperse dyes [1].

Depending very much on the final applications, many different types of nanomaterials can be used in polyolefins. They are carbon nanotubes (CNTs), nanosized silicates and silica, nanoceramics, nanometals, to name just a few. The nanomaterials can take different forms, being zero dimensional, such as nanospherical silica, one-dimensional, such as nanowires and nanotubes, or two-dimensional, such as exfoliated nanoclays and graphene sheets [2]. One of many significant advantages of the nanocomposites over the conventional composites is that the nanomaterial's loading is low, most of the time being below 10%. This is because of the high specific surface area of the nanoparticles which, when embedded uniformly, allows them to have an excellent interaction with the host polymer molecules. A low loading means less change of the conventional properties while new properties are introduced to the nanocomposite, or better performance is obtained with lower additional cost.

11.2 Polyolefin nanocomposite fibres and films

11.2.1 *Nanoparticles used in producing nanocomposite fibres and films*

11.2.1.1 *Carbon nanotubes*

CNTs can be divided into two main groups: single-walled (SWNTs) and multiwalled (MWNTs). The SWNTs have a tubular structure with carbon atoms connecting with

each other through an sp [2] hybrid orbital to form the wall which has the thickness of one carbon atom. The Gaussian distribution of diameters of SWNTs is on the order of a nanometer (nm) [3]. The MWNTs have a concentrically cylindrical structure, or a structure like a rolled-up paper. The length of CNTs is typically in the range of micrometers, though 4-cm-long SWNT has been reported [4].

CNTs have excellent mechanical properties [5]. Depending on synthesis methods and measurement techniques, CNTs can have tensile strength in the range of 1–20 GPa and Young's modulus 0.4–4.15 TPa. However, most CNTs used in many composites show reduction in the properties up to one to two orders of magnitude due to defects. CNTs can be semiconductors or conductors depending on the diameter or the geometric configuration of the nanotube. CNTs have been widely used for reinforcing polymers [6]. Enhanced stiffness, thermal stability, and gas barrier properties can be obtained from PP carbon nanoparticle composite [7]. Low shrinkage (<5%) at high temperature (160°C) was obtained on the PP CNT composite fibre [8].

11.2.1.2 *Silicates and silicas*

The most common silicates are based on (SiO_4) tetrahedron anionic group. Its ligand can also contain other electronegative atoms, such as F. The balanced charges are often Na^+ , K^+ , Ca^{++} , Mg^{++} , Fe^{++} , Al^{+++} , etc. Silicates are the most abundant minerals, being in different chemical compositions from many sources. Clay minerals as one group are hydrous aluminum phyllosilicates, available naturally or synthetically. Montmorillonite is one of the clays available naturally, pertaining to the smectic subgroup of clay. The phyllosilicate nanoparticles have a layered structure, in the case of montmorillonite, a 2:1 sandwiched structure of two layers of SiO_4 tetrahedron wrapping one layer of O, OH, Al, and Mg octahedron coordinated by other cations. During nanocomposite synthesis, the layered structure can be intercalated or exfoliated depending very much on the conditions used.

Silica is a general term for silicon dioxide, SiO_2 , also available naturally or synthetically. Sometimes, silica is considered a type of silicate. Synthetic silica can be in different physical forms, the most used one in nanocomposites being spherical porous mesostructure. Silicate and silica nanoparticles are either reengineered natural materials or synthesized materials on the nanometer scale, with an appropriate surface to prevent agglomeration. It has been reported [9] that the PP montmorillonite nanocomposites can have improved tensile characteristics, higher heat deflection temperature, retained optical clarity, high barrier properties, better scratch resistance, and increased flame retardancy. Also, improved strength, modulus, and elongation at break of PP silica nanocomposites can be achieved [10].

11.2.1.3 *Metal oxides*

Metal oxides, also known as ceramics, can be divided into semiconductors, such as ZnO, NiO, Fe_2O_3 , Cr_2O_3 , and TiO_2 , and insulators, such as MgO, CaO, and Al_2O_3 [11]. Due to the small size of the nanometal oxides, their surface possesses high

energy, especially for those materials with highly ionic nature, defects sites such as edges, corners, anion, and cation vacancies making the nanoparticles in this group very reactive, and thus valuable in nanocomposite fabrications.

In bulk, these metal oxides (ceramics) are brittle. However, their counterparts in nanoscale can be made more flexible, less brittle, and perhaps stronger [12]. The other properties of the nanometal oxides are also different. The bandgap of TiO_2 is in the UV range, 3.03 eV for rutile and 3.18 eV for anatase [13]. A blue shift can be observed as particle size decreases. Accordingly, it is widely studied for its photocatalytic capability. The photooxidative degradation of CHCl_3 with TiO_2 in water [14], however, indicated that when the nanoparticles become too small, the electron–hole recombination rate would be very fast leading to a lower efficiency in the initial light absorption step. The insulating metal oxides can also catalyze acid–base reactions, e.g., oxidative coupling of methane by MgO/CaO [15]. Hydroxyapatite reinforced PP was studied to be used as a bone analogue biomaterial [16]. $\text{SrTiO}_3/\text{NiZn}$ was used to improve the dielectric and magnetic properties of the prepared PP composite [17]. By using a specifically prepared ceramic, $\text{Y}_1\text{Ba}_2\text{Cu}_3\text{O}_{7-x}$, the physical–mechanical, superconducting, thermophysical properties of isotactic PP (iPP) ceramic composite were enhanced [18]. Special optical properties can be obtained on PP ZnO film, e.g., its blocking UV while transmitting visible light [19].

11.2.1.4 Metals

One of the useful forms of nanometals is colloids as giant clusters that have a size up to 100 nm [20], stabilized by suitable ligand or solvent. Metal nanowires are reported as Cr resist patterns on PET films [21], CoNi alloy [22], and different structures including wires, plates, and flowers [23]. Many metal nanoparticles are deposited as special coatings [24]. Some are used for catalytic applications [25]. Nanosilver deposited on PP nonwoven shows antibacterial function [26]. Silver/PP nanocomposite films showed not only improved thermal stability but also increased mechanical properties compared to neat PP film. Tensile properties of PP nanocomposites were largely improved compared with neat PP resin, and elongation also increased by 175% for the nanocomposites containing 1000 ppm silver nanoparticles [27]. The uniformly dispersed Pd nanoparticles can retard the thermal decomposition of PP [28]. A unique method of introducing nanometal particles to PP allows for the fabrication of large amounts (kilogram-scale) of polymeric nanoparticle composites. It is done through thermal decomposition of metal containing compounds (MRn ; $\text{M} = \text{Cr}, \text{Mo}, \text{W}, \text{Ti}, \text{Zr}, \text{Fe}, \text{Co}, \text{Ni}, \text{Pd}, \text{Pt}, \text{Cu}$; $\text{R} = \text{CO}, \text{HCOO}, \text{CH}_3\text{COO}, \text{C}_6\text{H}_5\text{CH}_2$) in solution melt of polymer, such as PP [29].

11.2.1.5 Nanostructured molecules

Two nanostructured molecules have been found useful for PP nanocomposites. One is dendrimers and the other cyclodextrins (CDs). Dendrimers are regularly branched polymers in which all bonds converge to a core, each repeat unit contains a branch junction forming the interior, and a large number of identical terminal functional

groups are present on the surface [30]. Dendrimers are usually viscous liquids at room temperature. The large amount of functional groups on the surface of a globularly shaped dendrimer molecule offer many attractive possibilities to modify other polymers, e.g., solubility, compatibility with plastics, uptake of guest molecules, and surface activity. The use of dendrimers and hyperbranched polymer as dyeing promoters for PP fibres was reported [31–33].

CDs are cyclic oligosaccharides formed by 6 (α -CD), 7 (β -CD), or 8 (γ -CD) α -1,4-D-glucopyranoside units resulting in a toroid structure with interior hydrophobic cavities of 0.5–1.0 nm in diameter and exterior hydrophilic rims. Due to the unique structure, CD can be used to form host–guest complexes for many applications. iPP coalesced in γ -CD and then released from the host showed different characteristics [34]. The crystalline inclusion complex between β -CD and low-molecular-weight PP was obtained and investigated. α -CD and γ -CD did not form crystalline inclusion complexes with PP [35]. Grafting of CDs onto PP nonwoven fabrics for the manufacture of reactive filters was reported [36].

11.3 Preparation of polyolefin nanocomposite fibres and films

11.3.1 Solution mixing

PP/montmorillonite nanocomposites were prepared in our laboratory by solution mixing followed by melt mixing and/or melt spinning. Ball milled nanoclay and 400 mL of xylene were added to a stainless steel container and this container was placed in a sand bath insulated with glass fabric. Various clay concentrations were added, namely 2%, 4%, and 6% on the weight of polymer. The contents of the container were ultrasonicated for 1 hours. The ultrasonication parameters were 90% amplitude, 8-second pulse on, and 4-second pulse off. To the mixture (xylene + clay), 0.4 wt% (on the weight of the polymer) of titanate coupling agent was added and ultrasonication was continued for another hour. PP pellets were then added to the mixture. The system was heated on a hot plate and the temperature was gradually raised to the boiling point of xylene (140°C) to dissolve the PP pellets. Further ultrasonication was done to homogenize the mixture. After the homogenization process, the ultrasonic device was switched off and the transducer was removed. The remaining xylene was allowed to evaporate at its boiling point until the nanocomposite solidified. Subsequently, the nanocomposite was mechanically crumbled and dried in an oven at 70°C to remove any traces of xylene.

11.3.2 Melting, mixing, and spinning

The nanocomposites obtained from solution mixing were mixed in a Brabender-plasticorder melt mixer at 170°C for 30 minutes. The speed of the screw in the melt mixer was maintained at 70 rpm.

The melt spinning of solution mixed nanocomposites was carried out in a single screw Sterling Extruder. The fibre extrusion was performed using a 30-spinneret fibre die and the following extrusion parameters:

- Temperature of different zones: 163°C (hopper), 185°C, 185°C, 191°C, 191°C, and 191°C (spinneret)
- Screw speed: 8 rpm
- Material throughput: 4.8 g/min
- Spinneret diameter: 0.5 mm
- Winding speed: 31 m/min
- Draw ratio: 4
- Drawing temperature: 100°C.

11.4 Characterization and analysis

11.4.1 *Scanning electron microscopy*

The surface topology of samples can be explored using scanning electron microscopy (SEM). For SEM measurements in our laboratory, samples of dimensions 8 mm × 4 mm were prepared using a sharp knife. The samples were then mounted on aluminum holders in order to be coated with gold (Au). Coating used in SEM analysis is aimed mainly at the following:

1. Preventing the charge-up on the specimen surface by covering it with a conductive material.
2. Increasing secondary electron emission by covering a specimen of low secondary electron emission with a metal of high secondary electron yield.

For coating, the vacuum evaporation method and the sputtering method were used. Care was taken to control the amount of coating. If the coating is too thick, its particles become visible while at the same time the structures of interest may be obscured. The samples were coated using a sputtering device most widely used for observing specimen surface morphology.

The measurements were carried out on a JEOL JSM 5610 SEM. For higher magnifications (> 10,000) the accelerated voltage was set at 10 kV while for lower magnifications a voltage of 5 kV was applied. Similarly, the working distance (WD) for lower magnifications ranged between 50 and 55 mm and for higher magnifications it was between 10 and 15 mm. A focused electron beam with accelerating voltage from 0.1 to 50 kV was used to scan across the sample surface. At the incident point of the primary electron beam, secondary electrons are emitted. The intensity of the secondary electrons depends on the surface topography from which the size and size distribution of nanoparticles can be characterized.

11.4.2 *Transmission electron microscopy*

Transmission electron microscopy (TEM) is a powerful technique used to study structures at and below the nanometer scale. Therefore it can be used to confirm

results obtained by X-ray diffraction (XRD) about the organization of nanoparticles in the nanocomposite. It allows a precise observation of nanostructures with exceptional resolution (about 0.2 nm). This technique is widely used by researchers worldwide to characterize polymer nanocomposites.

Manias et al.[9] reported that although XRD could give a fast answer on the miscibility of polymer/silicate systems, it did not give all the structural information of the nanocomposites. It was also reported that XRD provided no information about any exfoliated or disordered clay layers since they possessed no periodic structure. However, TEM was found useful in observing the nanocomposite structure. It provided the information about the existence of a substantial amount of intercalated clay layers throughout the PP matrix, in arrangements of single, double, and three layers [37]. Though there are some disadvantages, both XRD and TEM are considered to be the most useful techniques for characterizing polymer nanocomposites. In TEM, the specimen regions which are thicker or of higher density scatter the electron beam more strongly and will appear darker in the image. The nanoclay has a higher density than the semicrystalline PP (about 2.40 and 0.90, respectively). Therefore, the clay layers will appear darker than the polymer. If the specimen is too thick, the polymer appears darker and this reduces the contrast with the clay. This explains why the specimen should be cut as thin as possible. A high-resolution transmission electron microscope, JEOL 2010F FasTEM operating at 200 kV, with a point resolution of 1.9 Å and lattice resolution of 1.4 Å, was used to observe the physical state of clay particles in the nanocomposites. For TEM sample preparation, a diamond blade was used to scratch the small specimen pieces from the bulk sample. These small pieces were then ground in an agate mortar with acetone. The acetone suspension was then pipetted onto a carbon-coated Cu grid. This prepared sample was used for TEM observation.

11.4.3 X-ray diffraction

In addition to errors connected to the experimental setup, the XRD analysis of polymer–clay nanocomposites is further complicated by two factors. First, nanocomposites generally contain a fairly small amount of clay (typically <10 wt%). Therefore, the XRD analysis must be sensitive enough to detect the crystalline structure of the clay in the polymer. If this is not achieved, no peaks appear in the diffraction pattern and a false conclusion that a delaminated nanocomposite has been synthesized might be drawn. Moreover, the depth of penetration of X-rays is inversely proportional to the diffraction angle θ ; this means that X-ray analysis at a low angle only reflects the structure present in a thin layer close to the surface (typically 0.1 mm for polymers). Therefore, a thin sample with a large surface area is recommended for analysis. Wide-angle X-ray scattering was conducted at ambient temperature on a rotating anode diffractometer, Rigaku 18KW, with Cu K α radiation of wavelength 1.54 Å. The accelerating voltage was 60 kV. Montmorillonite clay particles were studied as powders and the nanocomposites were studied as 400 μ m thin films. XRD data were used to determine the d-spacing and the crystallite thickness of the clay particles.

11.4.4 Fourier transform infrared spectroscopy

Almost any compound having covalent bonds, whether organic or inorganic, can be found to absorb electromagnetic radiation in the infrared region of the spectrum, which lies at wavelengths longer than visible light (400–700 nm) but shorter than microwaves (longer than 1 mm). For characterization purposes, attention is paid to the vibrational portion of the infrared region located at wavelengths between 2.5 and 25 μm or in more common terms, frequencies between 4000 and 400 wave numbers. With the advent of computerized Fourier transform infrared (FTIR) spectrometers, which provide the concomitant advantages of speed, sensitivity and higher resolution, infrared spectroscopy has become one of the preeminent tools for molecular characterization and identification.

Some representative IR absorption peaks of crystalline PP are given in Table 11.1 [38]. The CH vibrational modes of PP are tabulated in Table 11.2 [39].

However, even in the case of FTIR, it may be difficult to exactly determine wavelength and intensity of a given absorption due to either the overlap of interfering bands or appearance of a strong background absorbance. Such problems can often be solved by using derivative data processing. The second derivative mode provides the advantages of improving the resolution of overlapping bands, suppression of the background to correct for systematic error, and enhancement of fine spectral features for qualitative and quantitative analysis.

The general form of an IR spectrum can be considered as a combination of a series of Gaussian distribution functions:

$$A = A_0 \exp(-Z^2/2\sigma^2)$$

where A = absorbance at wavelength λ , A_0 = absorbance at λ_{max} which is the wavelength of maximum absorption, $Z = (\lambda - \lambda_{\text{max}})$, and σ = standard deviation; and Lorentzian distribution functions [39]:

$$A = A_0(1 + Z^2/3\sigma^2)^{-1}$$

Table 11.1 IR absorption peak assignment

Frequency (cm^{-1})	Assignment
805, 840, 898, 940, 972, 995, 1044, 1100, 1152, 1165	CH ₂ rocking, C–C stretching, and CH ₃ wagging and rocking
1218, 1255, 1293	CH ₂ twisting and CH bending
1303, 1330, 1360	CH ₂ wagging and CH bending
1380	Symmetric CH ₃ bending
1440, 1455, 1465	CH ₂ bending and antisymmetric CH ₃ bending
2866, 2875, 2907, 2918, 2926	Symmetric CH ₃ stretching, antisymmetric CH ₂ stretching, and CH stretching
2947	Antisymmetric CH ₃ stretching
2960	Antisymmetric CH ₃ stretching

Table 11.2 Classification of the CH vibration modes of PP

CH mode	CH ₂ mode	CH ₃ mode
Stretching Wagging Rocking	Symmetric stretching Antisymmetric stretching Bending Wagging Rocking Twisting	Symmetric stretching Antisymmetric stretching 1 Antisymmetric stretching 2 Symmetric bending Antisymmetric bending 1 Antisymmetric bending 2 Wagging Rocking Torsion

For a given absorbance spectrum, the first derivative ($dA/d\lambda$, A is the absorbance and λ the wavelength) is the absorbance gradient of the original band at each wavelength. Repeated differentiation produces the second and higher derivatives:

$$\frac{d^2A}{d\lambda^2} \cdots \frac{d^nA}{d\lambda^n}$$

A family of such curves is illustrated in Fig. 11.1. The first derivative is a bilobal curve, intercepting the wavelength axis at the wavelength of the maximum absorbance ($\lambda_{\max}$) of the corresponding peak in the original zero-order IR spectrum. The second derivative curve is seen as an inverted main (centroid) peak flanked by two smaller satellite peaks. This pattern is repeated for each of the higher derivatives, the odd derivatives maintaining a zero crossing at the peak position of the original band ($\lambda_{\max}$) and the even derivatives a centroid band, the peak of which corresponds to $\lambda_{\max}$ of the original band. The even derivative centroid peak alternates in sign and decreases in width with increasing derivative order. Thus as derivative order increases, fine spectral detail is enhanced, so that spectral profiles are obtained in which overlapping bands become progressively more resolved.

Several sampling methods including conventional potassium bromide (KBr) disk transmittance, attenuated total internal reflectance, and diffuse reflectance can be applied. The general measurement conditions are as follows:

- Scan speed: 0.4 cm/s
- Scanning number: 64
- Resolution: 4 cm⁻¹
- Temperature: ambient temperature.

11.4.5 Thermal analysis

Incorporation of nanoparticles in the polymer melt usually has varying effects on the crystallization. Differential scanning calorimetry (DSC) studies were carried out in our laboratory on the neat PP and nanocomposite fibres in order to determine

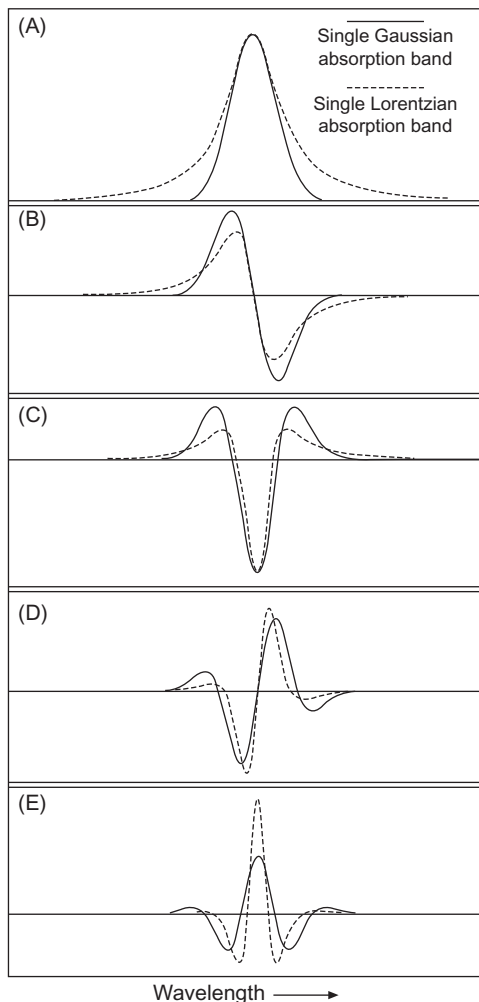


Figure 11.1 Derivative modes of absorption bands: (A) zero order, (B) first derivative, (C) second derivative, (D) third derivative, and (E) fourth derivative.

any change in percentage crystallinity, melt peak temperature, and melt onset temperature of nanocomposites fibres as compared to those of neat PP.

DSC analysis was carried out using nitrogen gas to provide an inert atmosphere. Specimens weighing not more than 10 mg were weighed to an accuracy of 0.01 mg into clean dry aluminum specimen pans specially designed for DSC apparatus. The sample pan and an empty reference pan made of the same material were then loaded into the test chamber purged with dry nitrogen gas at a constant flow rate of 10–50 mL/min during the experiment. The sample pan and reference pan were then heated to 220°C at the rate of 10°C/min. The instrument measures the

additional heat flow required to maintain the sample pan at the same temperature as the reference pan. The result is depicted as a plot of temperature vs time usually consisting of exothermic or endothermic peaks. After the experiment, software (Universal Analysis) provided by TA Instruments was used to calculate the heat of fusion by integrating the area under the melting peak. The software provides the percentage crystallinity from the ratio of heat of fusion calculated from the plot obtained and the heat of fusion for the polymer crystal expressed as a percentage. The heat of fusion of PP crystal was taken as 207 J/g. Other characteristics such as melt onset temperature and melt peak temperature were obtained by analyzing the calorimetric plot with the help of software tools.

11.5 Applications of polyolefin nanocomposites

11.5.1 Physical applications

11.5.1.1 Mechanical properties

Polyolefin boehmite nanocomposites showed increased stiffness with increasing the boehmite aspect ratio and accompanied by an increase in elongation at break [40]. Thermoplastic vulcanizate (TPV)/montmorillonite nanocomposite based on PP ethylene–propylene rubber (EPDM), and organically modified montmorillonite (OMMT) was prepared. It was found that the increase in the tensile strength of the TPVOMMT nanocomposite was 55% vs TPV, and 33% vs the dynamic vulcanizate that was based on PP/EPDM/Na-MMT (TPVNMMT). Dynamic mechanical analysis showed that the storage modulus of the TPVOMMT nanocomposite increased by 35% compared with that of TPV, while T_g of TPVOMMT increased by 8°C. Complex viscosity decreased with the addition of OMMT [41]. A PP with organo-modified clays (2 wt%) and different coupling agents showed a 25%–50% increase in tensile strength. Essential work of fracture was improved by 20% [42]. Full exfoliation of clay/high-density polyethylene (HDPE) nanocomposites can be achieved with the aid of maleated HDPE (polyethylene-grafted maleic anhydride, PE-g-MAN). A modest increase from 20.8 to 21.7 MPa in yield strength and an increase from 213 to 254 MPa in tensile modulus of the nanocomposite with 0.05% clay content over the base material were observed. The values level off above this clay loading and exhibit a further increase when the clay content is higher than 1 wt%. The yield strain decreased with increasing clay content [43]. The fracture properties of thermoplastic olefin 5% nanosilica composites were improved compared to the neat thermoplastic olefin blend. The fracture modulus was 920 MPa and the flexural stress was 28.4 MPa for the nanocomposite while the neat one was 797 and 25.5 MPa, respectively [44].

11.5.1.2 Gas permeability

Oxygen transmission rate (OTR in cc/m² day) of polyolefin nanoclay composites can be reduced. The composition of the samples was 90% PP or LDPE, 5% MBPP, or MBLDPE (master batch containing maleic anhydride), 5% Cloisite-15A

nanoclay. The PP nanocomposite film showed a reduced OTR at 374 down from 480 for the neat PP film. The LDPE nanocomposite film showed a similar trend being at 210 from 240 for the neat LDPE [45]. The oxygen permeability (OP, mol O₂/m s Pa) of paraffin nanoclay films dramatically decreased with increasing clay percentage except for the highest loading at 15 wt%. With only 5 wt% organoclay, the OP dropped by approximately 62 times (from 4629×10^{-17} to 75×10^{-17}). Increasing the clay loading to 10 wt% reduced the OP by a factor of 330 times (from 4629×10^{-17} to 14×10^{-17}) relative to the clay-free system. However, continued increase in the organoclay concentration led to embrittlement of the film, accompanied by a complete loss of any barrier improvement [46].

11.5.1.3 Thermal properties

The linear thermal expansion of maleated PP organoclay nanocomposites can be significantly reduced in the flow and transverse directions of injection-molded specimens because of the combined effect of the mechanical constraint by organoclay and the highly elongated elastomer particles caused at high-polypropylene-grafted maleic anhydride (PP-g-MA) contents [47]. CaCO₃ nanoparticle shapes (spherical and elongated) can affect the thermal properties of iPP. The iPP/CaCO₃ nanocomposites showed both increased decomposition temperature (378–391°C, vs iPP 368°C) and glass transition temperature (22–26°C, vs iPP 20°C) depending on the shape and add-on of the CaCO₃ used [48]. Intumescent PP compounds containing 1.5% silicone elastomer and 0.5% montmorillonite nanoparticles reached V0 UL 94 rating (burn <10 s without dripping). It was proposed that the silicone-elastomer-coated nanoparticles accumulate on the surface in fire and act as a skeleton for a barrier layer. The silicone elastomer may act as glue keeping nanoparticles in the protecting surface layer together at high temperature [49]. A norbornylene-substituted polyhedral oligomeric silsesquioxane (POSS) macromonomer was used to form polyolefin POSS nanocomposites [50]. Up to 56 and 73 wt% POSS can be achieved in PE and PP systems, respectively. The decomposition onset temperature based upon 5% mass loss was increased by 90°C, while the tensile properties were maintained for the PE POSS system. PP/corn starch master batch/modified montmorillonite composites with 1, 3, 5, and 7 wt% montmorillonite were prepared. An increase in degradation temperature with the montmorillonite amount was observed [51]. The exfoliated clay/HDPE nanocomposites were compared with their intercalated counterparts. A reduction in the burning rate from 25 to 21 mm/min for exfoliated HDPE nanocomposites and from 25.3 to 22.8 mm/min for the intercalated sample at clay content 1.0 wt% was observed [43]. It was found that the decomposition temperature of the PE nanocomposite with the organoclay of 11.91 wt% could be 46.8°C higher than that of pure PE when 30% weight loss was selected as a measuring point, showing enhanced thermal oxidation stability of this kind of polyolefin nanocomposite [52]. A 166°C increase in the onset degradation temperature ($T_d = 484^\circ\text{C}$) was achieved on a PP 5 wt% montmorillonite nanocomposite specifically made using an ionic liquid, 1-methyl-3-tetradecylimidazolium chloride, modified montmorillonite [53]. Synthetic hydromagnesite (HM) was used in

Table 11.3 Combustion parameters

Sample	LOI	Epiradiateur			Cone calorimeter ^a		
		No. of ignitions	Av. combustion extent (s)	Time to Ignite (TTI) (s)	PHRR (kW/m ²)	AHRR (kW/m ²)	FPI (m ² s/kW)
LDPE/EVA	18.7	1	∞	40	1135	304	0.035
ATH/HM	28.6	19	7.9	103	182	139	0.566
ATH/MH	28.1	19	7.8	70	180	112	0.389
ATH/U	27.1	18	8.4	71	181	111	0.392

Note: PHRR, peak heat release rate; AHRR, average heat release rate; FPI, fire performance index.

^aWith $\pm 10\%$ reproducibility.

Table 11.4 Smoke data

Sample	Av. SEA (m ² /kg)	CO/CO ₂ weight ratio	Av. CO emission (kg/kg)	Av. CO ₂ emission (kg/kg)	TSR
LDPE/EVA	201	0.009	0.015	1.70	827
ATH/HM	221	0.015	0.031	2.05	929
ATH/MH	69	0.018	0.022	1.20	225
ATH/U	102	0.006	0.012	2.18	368

Note: SEA, smoke extinction area; TSR, total smoke released.

combination with aluminum hydroxide (ATH) and compared with commercial flame retardants like magnesium hydroxide (MH) and natural HM huntite (U) in a polyolefin system of low-density polyethylene/poly(ethylene-*co*-vinyl acetate) (LDPE/EVA, 75/25). Combustion parameters obtained from limiting oxygen index (LOI), epiradiateur, and cone calorimeter are given in Table 11.3. The smoke data are given in Table 11.4 [54].

11.6 Chemical applications

11.6.1 Coloration

PP clay (montmorillonite) nanocomposites can be dyed as nylon with acid dyes and as polyester with disperse dyes. The colors obtained on the nanocomposite films were decent and comparable with that on the traditional textile fibres. The color build-up on the nanocomposite films is measured as a *K/S* value. The higher the *K/S*, the darker the color. Figs. 11.2–11.5 show the color strength of the PP clay nanocomposite films with 2%, 4%, and 6% nanoclay content.

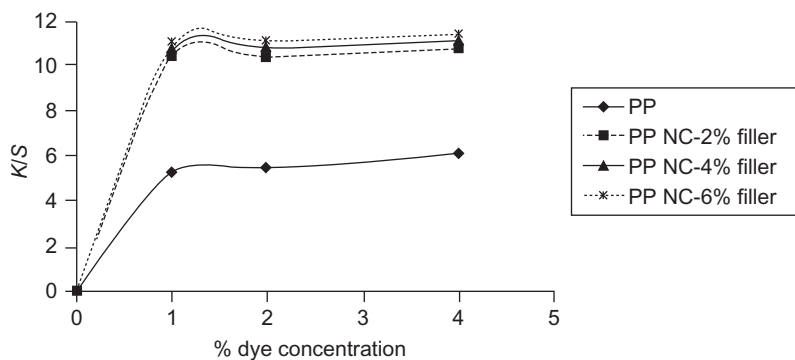


Figure 11.2 Color build-up curve with C. I. Acid Red 266.

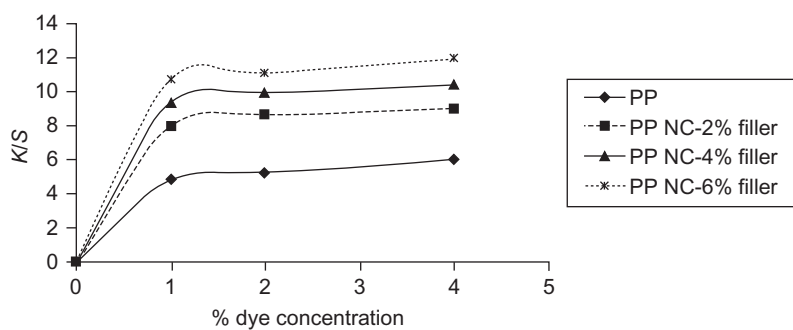


Figure 11.3 Color build-up curve with C. I. Acid Blue 113.

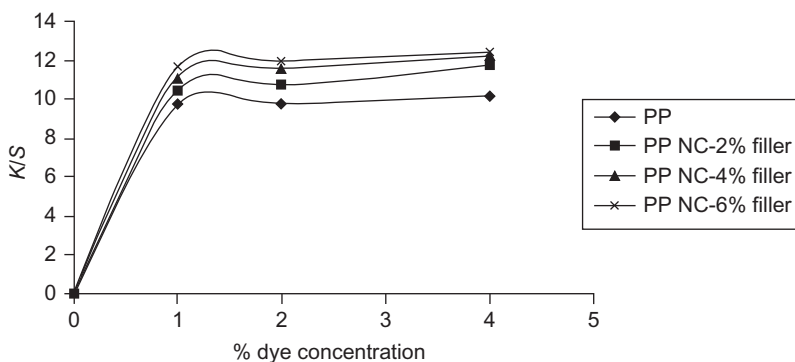


Figure 11.4 Color build-up curve with C. I. Disperse Red 65.

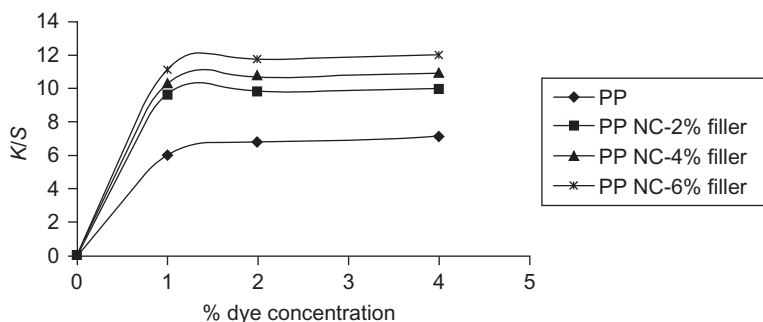


Figure 11.5 Color build-up curve with C. I. Disperse Yellow 42.

Table 11.5 Antibacterial activity of PP/clay nanocomposite films

Bacteria	Percentage reduction in bacterial number					
	2% nanoclay		4% nanoclay		6% nanoclay	
	Avg.	St. dev.	Avg.	St. dev.	Avg.	St. dev.
<i>Staphylococcus aureus</i>	99.94	0.03	99.78	0.1	99.95	0.04
<i>Klebsiella pneumoniae</i>	99.97	0.02	99.99	0.01	99.99	0.01

11.6.2 Antimicrobial property

With 2% nanoclay (montmorillonite) modified with dimethyl dihydrogenated tallow quaternary ammonium, the PP/montmorillonite nanocomposite films showed their ability to reduce the bacteria counts as compared to neat PP control films. The reduction in bacterial number for nanocomposite films was very close to 100% as given in Table 11.5.

However, it should be pointed out that the antibacterial activity of the PP/clay nanocomposites was found to be associated with the cationic nature of the clay modifying agent, the quaternary ammonium. At a processing temperature higher than 170°C, the antibacterial properties of the nanocomposites are lost due to the decomposition of the modifying agent. Table 11.6 provides the antibacterial activity of modified and unmodified montmorillonite after treatments at different temperatures. Therefore, it is recommended that the processing of PP/clay nanocomposites should be carried out at a temperature below the decomposition temperature of the cationic modifying agent, if used, in order to make the nanocomposites antibacterial.

11.6.3 Other applications

The 16-nm surface-hydroxylated SiO₂ homogeneously dispersed in PP resulted in a nanocomposite showing high transparency when SiO₂ add-on is above 2.5 wt% [55]. For PP nanoclay composites, the increased clay content from 0% to 10% can

Table 11.6 Antibacterial activity of modified and unmodified montmorillonite

Clay	Percentage reduction in bacterial number			
	<i>Staphylococcus aureus</i>		<i>Klebsiella pneumoniae</i>	
	St. dev.	Average	St. dev.	Average
Modified	98.18	1.76	96.12	1.04
Modified and treated at 170°C	97.93	1.75	95.86	1.13
Modified and treated at 191°C	31.49	0.81	28.51	1.43
Unmodified	14.71	1.67	14.17	0.81

lead to an increased contrast ratio from 29.35 to 56.63. The higher the contrast ratio, the more opaque the sample [56].

11.7 Conclusions

The use of nanomaterials has made modification of polyolefins easy and available for many applications. Strong and active research and development (R&D) of polyolefin nanocomposites have been manifest both in the literature and patents. However, it is worth noting that the performance of polyolefin nanocomposites is dependent upon not only the nanomaterials used, but also the polymer matrix, as well as processing conditions. Many factors in the fabrication can affect the properties of the resultant nanocomposites, the nanoparticles' size and size distribution, crystalline structure, surface modification, polyolefin tacticity, molecular weight (degree of polymerization), polymer crystallinity, mixing temperature, mixing time, etc., to name just a few. A uniform distribution of the nanoparticles inside the host polymer matrix is equally important. If sheet-like nanostructure is used, exfoliation can usually result in a better reinforcement than intercalation due to a better interaction of the host polyolefin molecules and the exfoliated nanomaterials. Though solution mixing and melt compounding for nanocomposite fabrication are presented in this chapter, there are other ways of fabricating the polyolefin nanocomposites. In situ polymerization based on metallocene methylaluminoxane catalysis [57–59] is one of many preparation approaches. Apart from the solid form of the nanocomposites, a porous form, foam, is also possible, which has a variety of applications in technical, commercial, and environmental areas. It was reported that foams with hectorite nanoparticles exhibit improved thermal stability and mechanical properties when compared to neat polymeric foams [60].

Functional polyolefin nanocomposites have widely extended the use of polyolefins. Some of the applications have been mentioned, but more appear in publications elsewhere, and even more will emerge as the R&D efforts in this field rapidly progress to attain the promise of improved performance for nanocomposites in packaging and engineering applications.

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